

Probability

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1) Random Experiment.

If the outcome is not unique but may be ^{any} one of the possible outcomes, the experiment is called a random exp.

Ex:- tossing a coin
throwing a dice

2) Outcome

the result of a random experiment is called an outcome.

Ex:- A coin is tossed; outcomes - head or tail.
A dice is thrown; outcomes - 1, 2, 3, 4, 5 or 6.

3) Trial and Event

Any particular performance of a random experiment is called a trial & outcome.

A combination of outcomes is called an event.

Ex:- tossing a coin is trial, getting a head is event.
throwing a dice is trial, getting 1, 2, 3, 4, 5, 6 is event.

4) Exhaustive Event:

the total no. of possible outcomes of a random exp. is called exhaustive event.

Ex:- In tossing a coin, there are 2 exhaustive events, head & tail.

• In throwing a dice, there are 6 exhaustive events, 1, 2, 3, 4, 5, 6.

5) Mutually Exclusive Events

Events are said to be mutually exclusive if the occurrence of one of them precludes the occurrence of all others in same trial. i.e., they cannot occur simultaneously.

Ex:- In tossing a coin, events head or tail are mutually exclusive ~~ex~~ since both head & tail cannot occur at same time.

6) Equally likely events

if the occurrence of none of them is expected in preference to others.

Ex:- In tossing a coin, head or tail are equally likely events.

In throwing a die, all the six faces are equally likely events.

7) Independent events

if occurrence of an event does not have any effect on occurrence of other event.

Ex:- In tossing a coin, event of getting a head in the first toss is independent of getting a head in second, third & subsequent tosses.

8) Favourable events

no. of outcomes which entail the occurrence of event

Ex:- In throwing a die, P.C of getting sum 5 is (1,4) (4,1) (2,3) (3,2)

9) Sample Space

A set of all possible outcomes is called sample space. Each element of the set is called a sample point or sample event.

10) Event

Any subset of a sample space is called event.

$$P(A) = \frac{\text{no. of favourable outcomes}}{\text{no. of total outcomes}} = \frac{m}{n}$$

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

$$P(A) \geq 0$$

$$P(S) = 1$$

$$P(A_1 \cup A_2 \cup A_3 \dots \cup A_n) = \sum_{i=1}^n P(A_i) \quad \{A_1, A_2, \dots, A_n = \text{mutually exclusive events}\}$$

$$P(\phi) = 0, \phi = \text{null set}$$

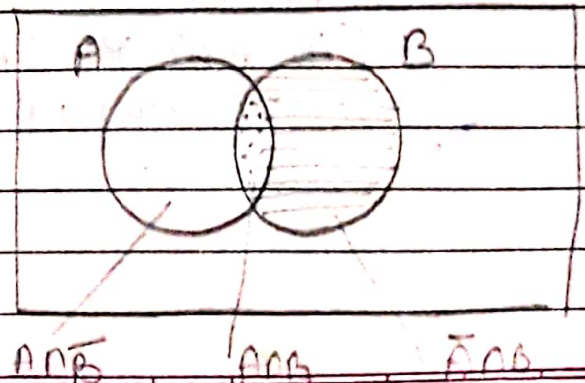
$$P(\bar{A}) = 1 - P(A)$$

$$A \cup \bar{A} = S \neq A \cap \bar{A} = \phi \quad \{A, \bar{A} \text{ are mutually exclusive events}\}$$

$$P(\overline{A \cap B}) = P(\bar{A} \cup \bar{B})$$

$$P(\overline{A \cap B}) = P(\bar{A} \cup \bar{B})$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$



- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- $P(A \cup B) = P(A) + P(B)$ $\left\{ \begin{array}{l} \text{if } A, B \text{ are m.e.e. i.e. } (A \cap B) = \emptyset \\ \text{ie. } (A \cup B) = 0 \end{array} \right\}$

- $A \cup B \text{ (A or B)} = A + B$

- $A \cap B \text{ (A and B)} = A \cdot B$

- $P(A \cup B) = 1 - P(\bar{A} \cap \bar{B})$

- $P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A \cup B) - P(A \cap B)$

- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$

- Occurrence of at least one
 $P(A \cup B \cup C) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

- Occurrence of at least two

$$P[(A \cap B) \cup (B \cap C) \cup (A \cap C)] = P(A \cap B) + P(B \cap C) + P(A \cap C) - 2P(A \cap B \cap C)$$

- Occurrence of exactly one = (at least one) - (at least two)

$$P[(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)] = P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(B \cap C) - 2P(A \cap C) + 3P(A \cap B \cap C)$$

- Occurrence of exactly two

$$P[(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)] = P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$$

- Conditional Probability

$$P(A \cap B) = P(A) - P(B|A) \\ = P(B) - P(A|B)$$

- $P(A \cap B) = P(A) \cdot P(B) \quad \{ A, B = \text{independent events} \}$

- $A = (A \cap B) \cup (A \cap \bar{B})$

- $P(A \cap B \cap C) = P(A) P(B|A) P[C/(A \cap B)]$

- $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) \quad \{ A, B, C = \text{independent events} \}$

- $P(A \cup B \cup C) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) \quad \{ \text{occurrence of at least one} \}$

- $P(A \cup B \cup C) = 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \quad \{ A, B, C = \text{independent} \}$

- $P(A \cup B \cup C) = P(A) + P(B) + P(C) \quad \{ A, B, C = \text{mutually exclusive} \}$

- Baye's Theorem

$$P(A_i|B) = \frac{P(A_i) P(B|A_i)}{\sum_{i=1}^n P(A_i) P(B|A_i)}$$

Random Variable

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1) Discrete Random Variable
if it takes either finite or countably infinite values.

Ex:- no. of children in family
no. of stars in sky

2) Continuous Random Variable
if it takes any values in given interval.

Ex:- height of person in cm
weight of bag in kg.

• Probability Mass Function

$$P(x_i) \geq 0 \quad \forall i$$

$$\sum_{i=1}^{\infty} P(x_i) = 1$$

$$X = x_i \quad x_1 \quad x_2 \quad x_3 \quad \dots \quad x_n$$

$$P(X = x_i) \quad P(x_1) \quad P(x_2) \quad P(x_3) \quad \dots \quad P(x_n)$$

• Discrete distribution Function

$$F(x) = P(X \leq x) = \sum_{i=1}^n p(x_i)$$

$$X \quad x_1 \quad x_2 \quad \dots$$

$$F(x) \quad P(x_1) \quad P(x_1) + P(x_2) \quad \dots$$

Probability Density Function

$$P\left(x - \frac{1}{2} dx \leq X \leq x + \frac{1}{2} dx\right) = f(x) dx$$

Properties

$$f(x) \geq 0, -\infty < x < \infty$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$P(a < x < b) = \int_a^b f(x) dx$$

Continuous distribution function

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx, -\infty < x < \infty$$

Properties

$$F(-\infty) = 0$$

$$F(\infty) = 1$$

$$0 \leq F(x) \leq 1, -\infty < x < \infty$$

$$P(a < x < b) = F(b) - F(a)$$

$$F'(x) = \frac{d}{dx} F(x) = f(x), f(x) \geq 0$$

- Two-Dimensional Discrete Random Variable
- Joint Probability Mass Function

$$p_{xy}(x_i, y_j) = P(X=x_i, Y=y_j) \quad \forall (x_i, y_j) \in \{(x, y)\}$$

$$p_{xy}(x_i, y_j) = 0 \quad \text{otherwise}$$

Conditions

$$p_{xy}(x_i, y_j) \geq 0 \quad \forall i \text{ and } j$$

$$\sum_{j=1}^n \sum_{i=1}^m p_{xy}(x_i, y_j) = 1$$

- Cumulative distribution Function

$$F_{xy}(x, y) = \sum_{j=1}^n \sum_{i=1}^m p_{xy}(x_i, y_j) = \sum \sum p_{ij}$$

Properties

$$F(-\infty, y) = 0 = F(x, -\infty) \quad \& \quad F(\infty, \infty) = 1$$

$$P(a < X < b, Y \leq y) = F(b, y) - F(a, y)$$

$$P(X \leq x, c < Y < d) = F(x, d) - F(x, c)$$

$$P(a < X < b, c < Y < d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

• Marginal Probability function

$$\begin{aligned} p_x(x_i) &= P(X=x_i) \\ &= P(X=x_i, Y=y_1) + P(X=x_i, Y=y_2) + \dots + P(X=x_i, Y=y_m) \\ &= p_{i1} + p_{i2} + \dots + p_{ij} + \dots + p_{im} \\ &= \sum_{j=1}^m p_{ij} = \sum_{j=1}^m p(x_i, y_j) = p_{i*} \end{aligned}$$

$$\begin{aligned} p_y(y_j) &= P(Y=y_j) \\ &= \sum_{i=1}^n p_{ij} = \sum_{i=1}^n p(x_i, y_j) = p_{*j} \end{aligned}$$

• Conditional Probability Function

$$p_{x|y} = P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$p_{y|x} = P(Y=y | X=x) = \frac{P(X=x, Y=y)}{P(X=x)}$$

$$P(X=x_i, Y=y_j) = P(X=x_i) \cdot P(Y=y_j) \quad [X, Y = \text{Independent}]$$

- Two dimensional Continuous random Variables
- Joint Probability Density function

$$P\left\{x - \frac{1}{2}dx \leq x \leq x + \frac{1}{2}dx \text{ and } y - \frac{1}{2}dy \leq y \leq y + \frac{1}{2}dy\right\} = f(x,y) dx dy$$

Properties

$$f(x,y) \geq 0 \quad \forall x,y$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$P(a \leq x \leq b, c \leq y \leq d) = \int_c^d \int_a^b f(x,y) dx dy$$

- Cumulative Distribution Function

$$F_{xy}(x,y) = \int_{-\infty}^y \int_{-\infty}^x f(x,y) dx dy$$

- Marginal Probability Function

$$f_x(x) = \int_{-\infty}^{\infty} f(x,y) dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f(x,y) dx$$

- Conditional Probability Function

$$f(x|y) = \frac{f(x,y)}{f_y(y)}, \quad f(y|x) = \frac{f(x,y)}{f_x(x)}$$

$$f(x,y) = f_x(x) f_y(y) \quad [x,y = \text{inde...}]$$

Basic Statistics

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- Measures of Central Tendency

- Mean

$$\mu = E(X) = \sum_{i=1}^{\infty} x_i p(x_i) = \sum x p(x)$$

$$E[\phi(x)] = \sum_{i=1}^{\infty} \phi(x_i) p(x_i) = \sum \phi(x) p(x)$$

Properties

$$E(X+k) = E(X) + k$$

$$E(aX \pm b) = aE(X) \pm b$$

$$E(X+Y) = E(X) + E(Y)$$

$$E(XY) = E(X)E(Y) \quad (X, Y \text{ independent})$$

- Median

$$M = \frac{1}{2} (x_k + x_{k+1})$$

- Geometric Mean

$$\log G = \sum_{i=1}^{\infty} (\log x_i) p(x_i) = \sum (\log x) p(x)$$

- Harmonic Mean

$$\frac{1}{H} = \sum_{i=1}^{\infty} \frac{1}{x_i} p(x_i) = \sum \frac{1}{x} p(x)$$

- Measures of Dispersion

- Quantile Deviation

$$Q = \frac{1}{2} (Q_3 - Q_1)$$

- Mean Deviation

$$\begin{aligned} MD &= E \{ |x - \mu| \} \\ &= \sum_{i=1}^{\infty} |x_i - \mu| p(x_i) \\ &= \sum |x - \mu| p(x) \end{aligned}$$

- Standard Deviation

$$\begin{aligned} SD = \sigma &= \sqrt{\sum_{i=1}^{\infty} x_i^2 p(x_i) - \mu^2} \\ &= \sqrt{\sum (x^2) - \mu^2} \\ &= \sqrt{E(x^2) - [E(x)]^2} \end{aligned}$$

Variance

$$\text{Var}(X) = \sigma^2 = E(x^2) - [E(x)]^2$$

Properties

$$\text{Var}(k) = 0$$

$$\text{Var}(kX) = k^2 \text{Var}(X)$$

$$\text{Var}(X+k) = \text{Var}(X)$$

$$\text{Var}(aX+b) = a^2 \text{Var}(X)$$

- Moments

- Central Moments.

$$\begin{aligned}\mu_r &= E \{ (x - u)^r \} \\ &= \sum_{i=1}^{\infty} (x_i - u)^r p(x_i) \\ &= \sum (x - u)^r p(x)\end{aligned}$$

If frequency distribution is given, $n = \sum f$;

$$p(x_i) = \frac{\sum f_i}{N} \Rightarrow \mu_r = \frac{\sum_{i=1}^{\infty} f_i (x_i - u)^r}{N}$$

Properties

$$\mu_1 = 0$$

$$\mu_2 = \sigma^2 \text{ or } SD = \sigma = \pm \sqrt{\mu_2}$$

$\mu_3 > 0$, positively skewed

$\mu_3 < 0$, negatively skewed

$\mu_3 = 0$, symmetrical

$$\text{Skewness, } \beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\text{Kurtosis, } \beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\mu_1 = \mu_3 = \mu_5 = \dots = \mu_{2r+1} = 0 \text{ [symmetric]}$$

• Raw Moments.

$$\mu_r' = E \{ (x-a)^r \}$$

$$= \sum_{i=1}^{\infty} (x_i - a)^r p(x_i)$$

$$= \sum (x-a)^r p(x)$$

If frequency distribution is given;

$$n = \sum f \Rightarrow p(x_i) = \frac{\sum f_i}{n} \Rightarrow \mu_r' = \frac{\sum_{i=1}^{\infty} f_i (x_i - a)^r}{N}$$

• Relation between Central and Raw Moments.

$$\mu_1 = \mu_1' - \mu_1' = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_2' \mu_1' + 2(\mu_1')^3$$

$$\mu_4 = \mu_4' - 4\mu_3' \mu_1' + 6\mu_2' (\mu_1')^2 - 3(\mu_1')^4$$

$$\mu_1' = \mu - a$$

$$\mu_2' = \mu_2 + (\mu_1')^2$$

$$\mu_3' = \mu_3 + 3\mu_2 \mu_1' + (\mu_1')^3$$

$$\mu_4' = \mu_4 + 4\mu_3 \mu_1' + 6\mu_2 (\mu_1')^2 + (\mu_1')^4$$

- Skewness

- Measures of Skewness

Absolute skewness = Mean - Mode

- Karl Pearson's Coefficient of Skewness

$$Sk = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

$$= \frac{\text{Mean} - \text{Mode}}{\sigma}$$

$$= \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

Coefficient of skewness lies b/w -1 and 1
positively skewed distribution, $Sk > 0$
negatively skewed distribution, $Sk < 0$
symmetrical distribution, $Sk = 0$

- Kurtosis

$$\beta_1 = \frac{l_3^2}{l_2^3}$$

$$\beta_2 = \frac{l_4}{l_2^2} = \frac{l_4}{\sigma^4}$$

$\beta_2 > 3$ = leptokurtic curve

$\beta_2 < 3$ = platykurtic curve

- Measures of Statistics for Continuous Random Variables

- Mean

$$\mu = E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x) dx$$

- Median

$$\int_a^m f(x) dx = \int_m^b f(x) dx = \frac{1}{2}$$

- Mode

$$f'(x) = 0 \text{ \& } f''(x) < 0 \text{ \& } a < x < b$$

- Geometric Mean

$$\log G = \int_{-\infty}^{\infty} (\log x) f(x) dx$$

- Harmonic Mean

$$\frac{1}{H} = \int_{-\infty}^{\infty} \frac{1}{x} f(x) dx$$

- Mean Deviation

$$MD = \int_{-\infty}^{\infty} |x - \mu| f(x) dx$$

- Quantile Deviation

$$\int_{-\infty}^{\infty} f(x) dx = \frac{r}{4} \quad , r = 1, 2, 3, \dots$$

$$Q = \frac{1}{2} (Q_3 - Q_1)$$

- Standard Deviation

$$SD = \sqrt{\text{Var}(x)} = \sigma$$

- Variance

$$\text{Var}(x) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

- Moments

$$\mu_r = \int_{-\infty}^{\infty} (x - a)^r f(x) dx$$

$$\mu'_r = \int_{-\infty}^{\infty} (x - a)^r f(x) dx$$

When $a = 0$, $\mu'_r = r^{\text{th}}$ order simple moments

$$\mu'_r = \int_{-\infty}^{\infty} x^r f(x) dx$$

- Skewness

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

- Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$$

- Expected Values of Two Dimensional Random Variables.

- Continuous

$$E[g(x,y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y) dx dy$$

- Discrete

$$E[g(x,y)] = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} g(x_i, y_j) p_{ij}$$

$$= \sum_x \sum_y g(x,y) f(x,y)$$

Properties

$$E(x+y) = E(x) + E(y)$$

$$E(xy) = E(x) \cdot E(y) \quad [x, y \text{ independent RV}]$$

• Conditional Expectation & Conditional Variance.

- Continuous.

$$E\{g(x, y) | y = y\} = \int_{-\infty}^{\infty} g(x, y) f(x|y) dx$$

$$= \int_{-\infty}^{\infty} \frac{g(x, y) f(x, y)}{f_y(y)} dx$$

$$E(x | y = y) = \int_{-\infty}^{\infty} \frac{x f(x, y)}{f_y(y)} dx$$

$$E(y | x = x) = \int_{-\infty}^{\infty} \frac{y f(x, y)}{f_x(x)} dy$$

$$\text{Var}(x | y = y) = E\left[\{x - E(x | y = y)\}^2 | y = y\right]$$

$$\text{Var}(y | x = x) = E\left[\{y - E(y | x = x)\}^2 | x = x\right]$$

- Discrete

$$E\{g(x, y) | y = y_j\} = \sum_{i=1}^{\infty} g(x_i, y_j) P(x = x_i | y = y_j)$$

$$= \sum_{i=1}^{\infty} \frac{g(x_i, y_j) P(x = x_i, y = y_j)}{P(y = y_j)}$$

$$= \sum_{i=1}^{\infty} g(x_i, y_j) \frac{p_{ij}}{p_{+j}}$$

$$E(X|Y=y_j) = \sum_{i=1}^n x_i P(X=x_i | Y=y_j)$$

$$\text{Var}(X|Y=y_j) = E\left[\{X - E(X|Y=y_j)\}^2 | Y=y_j\right]$$

Properties

If X & Y are independent R.V.,

$$E(Y|X) = E(Y)$$

$$E(X|Y) = E(X)$$

$$E(XY) = E[X \cdot E(Y|X)]$$

$$E(X^2 Y^2) = E[X^2 \cdot E(Y^2|X)]$$

• Chebyshev's Inequality

$$P\{|X - \mu| \geq k\sigma\} \leq \frac{1}{k^2} \quad (\text{upper bound})$$

$$\text{or } P\{|X - \mu| < k\sigma\} > 1 - \frac{1}{k^2} \quad (\text{lower bound})$$

Correlation & Regression

• Types of Correlations.

• Positive and Negative correlations

- If both variables vary in same direction, the correlation is said to be positive.

Height (cm)	150	152	155	160	162
Weight (kg)	60	62	64	65	67

- If both variables vary in opposite direction, correlation is said to be negative.

Price (₹/unit)	10	8	6	5	4	1
Demand (units)	100	200	300	400	500	600

• Simple and Multiple Correlations

- When only two variables are studied, the relationship is described as simple correlation.

Ex:- Quantity of money & price level, demand and price, etc

- When more than two variables are studied, the relationship is described as multiple correlation.

Ex:- relationship of price, demand and supply of a commodity.

• Partial and total correlations

- When more than two variables are studied excluding some other variables, the result is termed as partial correlation.
- When more than two variables are studied without excluding any variables, the result is termed as total correlation.

• Linear and non-linear correlation

- If the ratio of change between two variables is constant, the correlation is said to be linear.

Milk (l)	5	10	15	20	25	30
Wage (₹)	2	4	6	8	10	12

- If the ratio of change between two variables is not constant, the correlation is said to be non-linear.

Advertising expense (₹ in lakhs)	3	6	9	12	15
Sales (₹ in lakhs)	10	12	15	15	16

• Methods of studying correlation

• Graphic methods

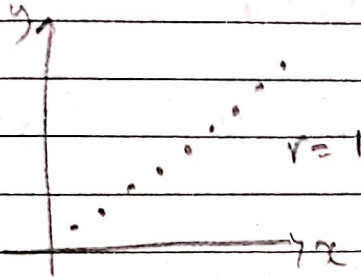
- Scatter diagram
- Simple graph

• Mathematical methods

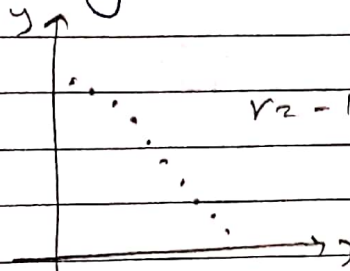
- Karl Pearson's coeff of correlation
- Spearman's Rank coeff of correlation

- Scatter Diagram.

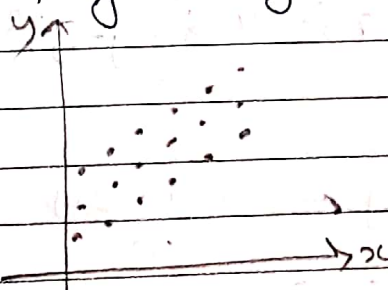
- Perfect Positive Correlation



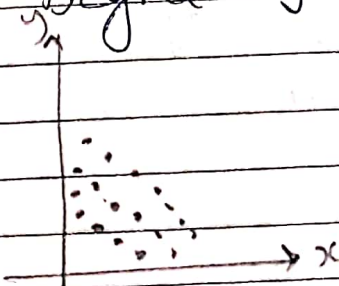
- Perfect Negative Correlation



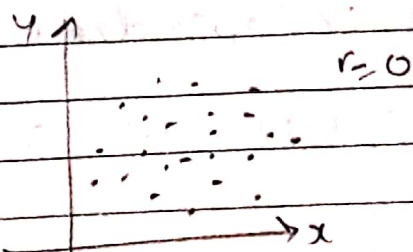
- High Degree of Positive Correlation



- High Degree of Negative Correlation



- No Correlation



- Karl Pearson's coeff. of Correlation

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{cov}(X, Y) = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y})$$

$$\sigma_X = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\sigma_Y = \sqrt{\frac{\sum (y - \bar{y})^2}{n}}$$

$$\therefore r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

$$= \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}} \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}}$$

Properties

$$\rightarrow -1 \leq r \leq 1$$

$\rightarrow$ If X, Y are independent,

$$\sum (x - \bar{x})(y - \bar{y}) = 0 \quad \text{or} \quad \text{cov}(X, Y) = 0$$

$$\therefore r = 0$$

• Rank Correlation

$$\sum (x - \bar{x})^2 = \frac{1}{12} (n^3 - n)$$

$$\sum (y - \bar{y})^2 = \frac{1}{12} (n^3 - n)$$

$$\text{If } d = x - y = (x - \bar{x}) - (y - \bar{y}) \quad [\because \bar{x} = \bar{y}]$$

$$\begin{aligned} \therefore \sum d^2 &= \sum [(x - \bar{x}) - (y - \bar{y})]^2 \\ &= \sum (x - \bar{x})^2 + \sum (y - \bar{y})^2 - 2 \sum (x - \bar{x})(y - \bar{y}) \end{aligned}$$

$$\sum (x - \bar{x})(y - \bar{y}) = \frac{1}{2} [\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2 - \sum d^2]$$

$$= \frac{1}{12} (n^3 - n) - \frac{1}{2} \sum d^2$$

$$\therefore r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

$$= \frac{\frac{1}{12} (n^3 - n) - \frac{1}{2} \sum d^2}{\frac{1}{12} (n^3 - n)}$$

$$= 1 - \frac{6 \sum d^2}{n^3 - n}$$

$$= 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$\sum d = \sum (x - y) = \sum x - \sum y = n(\bar{x} - \bar{y}) = 0$$

- Types of Regression

- Simple and Multiple Regression.

- The regression analysis for studying only two variables at a time is known as simple regression

- The regression analysis for studying more than two variables at a time is known as multiple regression.

- Linear and Non-linear Regression

- If the regression curve is a straight line, the regression is said to be linear.

- If the regression curve is not a st. line, the regression is said to be non-linear.

- Methods of studying regression

- Method of scatter diagram

- Method of least squares.

- Lines of Regression.

- Line regression of y on x

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y = a + bx$$

- line regression of x on y

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x = a + by$$

• Regression coefficients

b_{yx} = regression coefficient of y on x

$$= r \frac{\sigma_y}{\sigma_x}$$

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

b_{xy} = regression coefficient of x on y

$$= r \frac{\sigma_x}{\sigma_y}$$

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

Properties of regression coefficients

- $r = \sqrt{b_{yx} b_{xy}}$

- if $b_{yx} < 1$, $b_{xy} > 1$

- $\frac{1}{2} (b_{yx} + b_{xy}) \geq r$

- either both are positive or negative
- $r > 0$ if $b_{xy} > 0$ and $b_{yx} > 0$;
- $r < 0$ if $b_{xy} < 0$ and $b_{yx} < 0$.

• Properties of lines of regression.

- two regression lines X on Y & Y on X intersect at their means $(\bar{x}, \bar{y})$.
- $\because r = \sqrt{b_{yx} b_{xy}}$, $\therefore r, b_{yx}, b_{xy}$ all have ^{same} sign
- if $r = 0$, the regression coeff. are zero.
- the lines become identical if $r = \pm 1$.
- if $r = 0$, lines are $\perp$ to each other.

Some Special Probability Distributions.

- Binomial Distribution

$$P(X=x) = {}^n C_x p^x q^{n-x}, \quad p+q=1$$

p = prob. of success

q = prob. of failure

- Mean

$$E(X) = \sum x p(x)$$

$$= np$$

- Variance

$$\text{Var}(X) = \sum (x^2) p(x) - \mu^2$$

$$= npq$$

- Standard Deviation

$$SD = \sqrt{\text{Variance}}$$

$$= \sqrt{npq}$$

- Binomial Frequency Distribution

$$\sum_{x=0}^n f(x) = N \sum_{x=0}^n P(X=x) = N$$

$$\left[\because \sum_{x=0}^n P(X=x) = 1 \right]$$

- Poisson Distribution

$$P(X=x) = p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad , x=0, 1, 2, \dots$$

- Mean.

$$E(X) = \sum_{x=0}^{\infty} x p(x) \\ = \lambda$$

- Variance

$$\text{Var}(X) = E(X^2) - \mu^2 \\ = \lambda$$

- Standard Deviation

$$SD = \sqrt{\text{Variance}} \\ = \sqrt{\lambda}$$

- Exponential Distribution

$$f(x) = \lambda e^{-\lambda x} \quad , x > 0 \\ = 0 \quad , x \leq 0$$

where $\lambda > 0$ is called rate of distribution

- Mean

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \frac{1}{\lambda}$$

- Variance

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$= \frac{1}{\lambda^2}$$

- Standard Deviation

$$SD = \sqrt{\text{Var}(x)}$$

$$= \frac{1}{\lambda}$$

- Mode

$$x = 0$$

- Gamma Distribution

$$f(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, \quad x > 0$$

$$= 0, \quad x \leq 0$$

- Mean

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \frac{r}{\lambda}$$

- Variance

$$\text{Var}(x) = \frac{r}{\lambda^2}$$

- Standard Deviation

$$\text{SD} = \sqrt{\text{Var}(x)} = \frac{\sqrt{r}}{\lambda}$$

- Mode

$$x = \frac{r-1}{\lambda} \quad \text{if } r > 1$$

- Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$P(-\infty < x < \infty) = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1$$

$$P(-z_1 \leq z \leq 0) = P(0 \leq z \leq z_1) \quad , \quad z_1 > 0$$

- Mean = Median = Mode = μ

- Variance

$$\begin{aligned} \text{Var}(X) &= E(X - \mu)^2 \\ &= \sigma^2 \end{aligned}$$

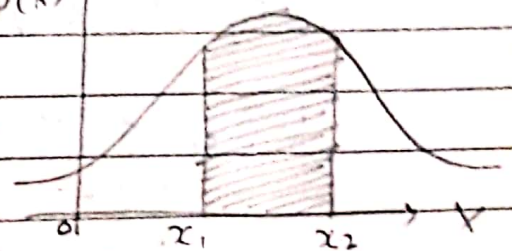
- Standard Deviation

$$SD = \sigma$$

- Probability of a Normal Random Variable

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{let } z = \frac{x - \mu}{\sigma}$$



$$E(Z) = E\left(\frac{x - \mu}{\sigma}\right) = \frac{1}{\sigma} [E(x) - \mu] = 0$$

$$\text{Var}(Z) = \text{Var}\left(\frac{x - \mu}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(x - \mu) = \frac{1}{\sigma^2} \text{Var}(x) = 1$$

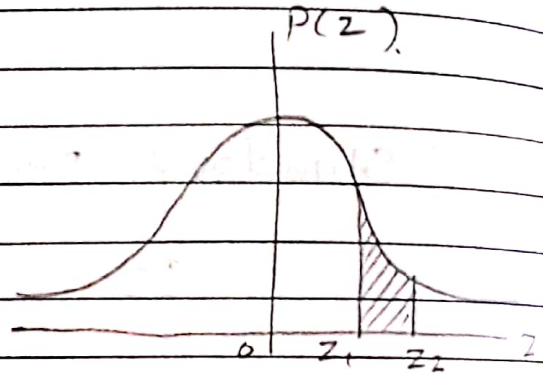
$$P(x_1 \leq X \leq x_2) = P\left[\left(\frac{x_1 - \mu}{\sigma}\right) \leq \left(\frac{X - \mu}{\sigma}\right) \leq \left(\frac{x_2 - \mu}{\sigma}\right)\right]$$

$$= P(z_1 \leq Z \leq z_2)$$

where $z_1 = \frac{x_1 - \mu}{\sigma}$ & $z_2 = \frac{x_2 - \mu}{\sigma}$

Case 1

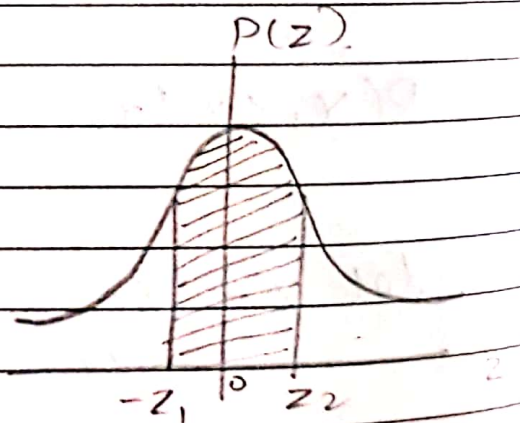
If $z_1 > 0$ or $z_1 < 0$
 $z_2 > 0$ $z_2 < 0$



$$\begin{aligned} P(x_1 \leq X \leq x_2) &= P(z_1 \leq Z \leq z_2) \\ &= P(0 \leq Z \leq z_2) - P(0 \leq Z \leq z_1) \end{aligned}$$

Case 2

If $z_1 < 0$ and $z_2 > 0$



$$\begin{aligned} P(x_1 \leq X \leq x_2) &= P(-z_1 \leq Z \leq z_2) \\ &= P(-z_1 \leq Z \leq 0) + P(0 \leq Z \leq z_2) \\ &= P(0 \leq Z \leq z_1) + P(0 \leq Z \leq z_2) \end{aligned}$$

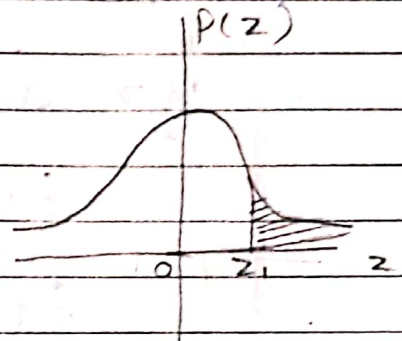
- When $x > x_1$, $z > z_1$. $P(z > z_1)$

Case 1

If $z_1 > 0$

$$P(x > x_1) = P(z > z_1)$$

$$= 0.5 - P(0 < z < z_1)$$



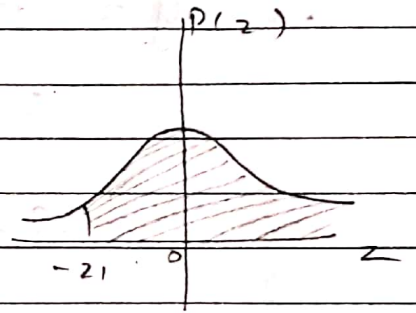
Case 2

If $z_1 < 0$

$$P(x > x_1) = P(z > -z_1)$$

$$= 0.5 + P(-z_1 < z < 0)$$

$$= 0.5 + P(0 < z < z_1)$$



- When $x < x_1$, $z < z_1$, $P(z < z_1)$

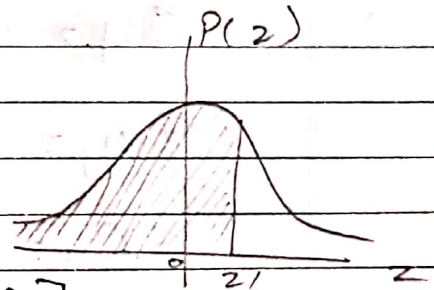
Case 1 If $z_1 > 0$

$$P(x < x_1) = P(z < z_1)$$

$$= 1 - P(z > z_1)$$

$$= 1 - [0.5 - P(0 < z < z_1)]$$

$$= 0.5 + P(0 < z < z_1)$$



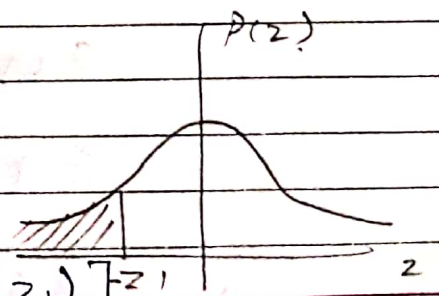
Case 2 If $z_1 < 0$

$$P(x < x_1) = P(z < -z_1)$$

$$= 1 - P(z > -z_1)$$

$$= 1 - [0.5 + P(0 < z < z_1)]$$

$$= 0.5 - P(0 < z < z_1)$$



Applied Statistics:

Test of Hypothesis.

- Procedure for testing of Hypothesis

- (i) Null Hypothesis (H_0)
- (ii) Alternative Hypothesis (H_1)
- (iii) level of significance (α)
- (iv) Test Statistic (z .)

$$Z = \frac{t - E(t)}{SE(t)}$$

- (v) Critical Value (Z_α)
- (vi) Decision, $|z| < Z_\alpha$ null hypothesis is accepted
 $|z| > Z_\alpha$ null hypothesis is rejected

- Test of Significance for large samples ($n > 30$)

$$Z = \frac{x - \mu}{\sigma}$$

if	$ z < 1.96$	not significant
	$ z > 1.96$	5% level of significance
	$ z > 2.58$	1% level of significance

• Test of Significance for Single Proportion Working Rule.

(i) Null Hypothesis H_0 ; $P = P_0$

(ii) alternative Hypothesis H_1 ; $P \neq P_0$
 $P > P_0$
 $P < P_0$

(iii) level of significance, α

(iv) Test Statistic,

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}, \quad Q = 1 - P$$

p = sample proportion

(v) Critical Value, Z_{α}

(vi) Decision, $|Z| < Z_{\alpha}$, null hypothesis is accepted
 $|Z| > Z_{\alpha}$, null hypothesis is rejected

Null Hypothesis is rejected when $|Z| > 3$
 without mentioning any level of
 significance.

- Test of significance for difference of proportions.

Working Rule.

(i) Null Hypothesis H_0 ; $P_1 = P_2$

(ii) alternative Hypothesis H_1 ; $P_1 \neq P_2$
 $P_1 > P_2$
 $P_1 < P_2$

(iii) level of significance α

(iv) Test Statistic

(a)
$$Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$
 When P_1 and P_2 are known

(b) (i)
$$Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$
 When P_1 & P_2 are not known

(ii)
$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

$$Z = \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

(v) critical value Z_α

(vi) Decision $|Z| < Z_\alpha$, accepted
 $|Z| > Z_\alpha$, rejected.

• Test of Significance for single mean

Working Rule

- (i) Null Hypothesis H_0 , $\mu = \mu_0$
- (ii) Alternative Hypothesis H_1 , $\mu \neq \mu_0$
- (iii) level of Significance α
- (iv) Test Statistic

(a) $Z = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$ When SD is known

(b) $Z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$ When SD is not known

(v) Critical value Z_α

(vi) Decision

$$\begin{array}{ll} |Z| < Z_\alpha & , \text{ accepted} \\ |Z| > Z_\alpha & , \text{ rejected.} \end{array}$$

- Test of Significance for difference of means

Working Rule

(i) Null Hypothesis H_0 ; $\mu_1 = \mu_2$

(ii) Alternative Hypothesis H_1 ; $\mu_1 \neq \mu_2$ (two tailed)
 $\mu_1 < \mu_2$ (one tailed)
 $\mu_1 > \mu_2$ (one tailed)

(iii) level of Significance α

(iv) Test Statistic

(a)
$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$
 When σ_1 & σ_2 are known

(b)
$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
 When not known

(v) Critical Value Z_α

(vi) Decision,

$$\begin{array}{ll} |Z| < Z_\alpha & , \text{ accepted} \\ |Z| > Z_\alpha & , \text{ rejected} \end{array}$$

- Test of Significance for difference of Standard Deviations

Working Rule,

(i) Null Hypothesis H_0 ; $\sigma_1 = \sigma_2$

(ii) Alternative Hypothesis H_1 , $\sigma_1 \neq \sigma_2$
 $\sigma_1 > \sigma_2$
 $\sigma_1 < \sigma_2$

(iii) Level of Significance α

(iv) Test Statistic

(a)
$$Z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$
 When σ_1 & σ_2 are known

(b)
$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
 When not known

(v) Critical Value Z_α

(vi) Decision,

$$\begin{array}{lcl} |Z| < Z_\alpha & , & \text{accepted} \\ |Z| > Z_\alpha & , & \text{rejected.} \end{array}$$

- Small Sample Tests ($n \leq 30$)
- Student's t-distribution

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} \quad \text{or} \quad t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)}$$

$\bar{x}$ = mean

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

- Test of Significance for single mean

Test Statistic,

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)} \quad \text{where } s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \quad \text{with } v = n-1$$

Critical value = $V = n-1$

- Test of Significance for difference of means

Test Statistic

$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where } V = n_1 + n_2 - 2$$

$$\bar{x} = \frac{\sum x}{n_1}$$

$$\bar{y} = \frac{\sum y}{n_2}$$

$$s^2 = \sqrt{\frac{\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2}{n_1 + n_2 - 2}}$$

If $n_1 = n_2 = n$

$$t = \frac{x - y}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}} \quad \text{df } 2n-2$$

- Test of significance for correlation coefficients

Test statistic

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad \text{df } n-2$$

- Snedecor's F-test for ratio of variances

$$F = \frac{s_1^2}{s_2^2} \quad , \quad s_1^2 > s_2^2$$

and, $s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$

$$s_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

and
$$s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1}$$

$$s_2^2 = \frac{\sum (y - \bar{y})^2}{n_2}$$

$$\Rightarrow S_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

F-test is right tailed test

$$F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$$

- Test of Significance for ratio of variances
Working Rule

- (i) Null Hypothesis $H_0: \sigma_1^2 = \sigma_2^2$
- (ii) Alternative Hypothesis $H_1: \sigma_1^2 \neq \sigma_2^2$
- (iii) level of significance α
- (iv) Test Statistics

$$F = \frac{S_1^2}{S_2^2}, \quad S_1^2 > S_2^2$$

(v) Critical Value, $\nu_1 = n_1 - 1$
 $\nu_2 = n_2 - 1$

(vi) Decision,

$F < F_\alpha$, accepted

$F > F_\alpha$, rejected.

- Chi-Square (χ^2) test

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

binomial distribution, $\nu = n - 1$

Poisson distribution, $\nu = n - 2$

normal distribution, $\nu = n - 3$

$$S.D = \sqrt{2\lambda}$$

- always positively skewed

- value lies b/w 0 and ∞

- Chi-Square test: goodness of fit

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Test of significance

Working Rule

- Null Hypothesis
- Alternative Hypothesis
- Level of Significance
- Calculate χ^2

$$\chi^2_{\alpha}$$

$$\begin{array}{l} \chi^2 < \chi^2_{\alpha} \quad , \text{ accepted} \\ \chi^2 > \chi^2_{\alpha} \quad , \text{ rejected} \end{array}$$

Curve Fitting

- Linear

- $$y = a + bx$$

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

- $$y = ax + b$$

$$\sum y = nb + a \sum x$$

$$\sum xy = b \sum x + a \sum x^2$$

- Quadratic

- $$y = a + bx + cx^2$$

$$\sum y = na + \sum bx + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

- $$y = ax^2 + bx + c$$

$$\sum y = nc + b \sum x + a \sum x^2$$

$$\sum xy = c \sum x + b \sum x^2 + a \sum x^3$$

$$\sum x^2 y = c \sum x^2 + b \sum x^3 + a \sum x^4$$

• Exp / Log.

$$y = ab^x \Rightarrow y = A + BX$$

$$\log y = \log a + x \log b$$

$$\log y = Y$$

$$\log a = A$$

$$x = X$$

$$\log b = B$$

$$y = ax^b \Rightarrow y = A + BX$$

$$\log y = \log a + b \log x$$

$$\log y = Y$$

$$\log a = A$$

$$b = B$$

$$\log x = X$$

$$y = ae^{bx} \Rightarrow y = A + BX$$

$$\log y = \log a + bx$$

$$\log y = Y$$

$$\log a = A$$

$$b = B$$

$$x = X$$